```
1000 'SUBROUTINE TO LOAD X1[ ] WITH IMAGES FROM THE DATABASE
1010 'Variables entering routine: LETTER%
1020 'Variables exiting routine: X1[1] to X1[100],  X1[101] = 1, CORRECT
1030 '
1040 'The variable, LETTER%, between 1 and 260, indicates which image in the database is to be
1050 'returned in X1[1] to X1[100].  The bias node, X1[101], always has a value of one.  The variable,
1060 'CORRECT, has a value of one if the image being returned is a vowel, and zero otherwise.
1070 '(The details of this subroutine are unimportant, and not listed here).
1900 RETURN


2000 'SUBROUTINE TO CALCULATE THE ERROR WITH THE CURRENT WEIGHTS
2010 'Variables entering routine:  X1[ ], X2[ ], WI[ , ], WH[ ], CORRECT
2020 'Variables exiting routine:   ELET
2030 '
2040 '                                          'FIND THE HIDDEN NODE VALUES, X2[ ]
2050 FOR H% = 1 TO 10                           'loop for each hidden nodes
2060   ACC = 0                                  'clear the accumulator
2070   FOR I% = 1 TO 101                        'weight and sum each input node
2080     ACC = ACC + X1[I%] * WH[H%,I%]
2090   NEXT I%
2100   X2[H%] = 1 / (1 + EXP(-ACC) )            'pass summed value through sigmoid
2110 NEXT H%
2120 '
2130 '                                          'FIND THE OUTPUT VALUE: X3
2140 ACC = 0                                    'clear the accumulator
2150 FOR H% = 1 TO 10                           'weight and sum each hidden node
2160   ACC = ACC + X2[H%] * WO[H%]
2170 NEXT H%
2180 X3 = 1 / (1 + EXP(-ACC) )                  'pass summed value through sigmoid
2190 '
2200 '                                          'FIND ERROR FOR THIS LETTER, ELET
2210 ELET = (CORRECT - X3)                      'find the error
2220 IF CORRECT = 1 THEN ELET = ELET*5          'give extra weight to targets
2230 '
2240 RETURN


3000 'SUBROUTINE TO FIND NEW WEIGHTS
3010 'Variables entering routine:  X1[ ], X2[ ], X3, WI[ , ], WH[ ], ELET, MU
3020 'Variables exiting routine:   WI[ , ], WH[ ]
3030 '
3040 '                                          'FIND NEW WEIGHTS FOR INPUT NODES
3050 FOR H% = 1 TO 10
3060   FOR I% = 1 TO 101
3070     SLOPEO = X3 * (1 - X3)
3080     SLOPEH = X2(H%) * (1 - X2[H%])
3090     DX3DW = X1[I%] * SLOPEH * WO[H%] * SLOPEO
3100     WH[H%,I%] = WH[H%,I%] + DX3DW * ELET * MU
3110   NEXT I%
3120 NEXT H%
3130 '
3140 '                                          'FIND NEW WEIGHTS FOR HIDDEN NODES
3150 FOR H% = 1 TO 10
3160   SLOPEO = X3 * (1 - X3)
3170   DX3DW = X2[H%] * SLOPEO
3180   WO[H%] = WO[H%] + DX3DW * ELET * MU
3190 NEXT H%
3200 '
3210 RETURN
```

TABLE 26-3